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# National Security Space Contributions to Operation Midnight Hammer

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This past weekend U.S. military forces conducted surprise attacks on various targets central to Iran's quest to achieve a nuclear weapons capability. According to Department of Defense (DoD) officials, the raid included seven U.S. Air Force (USAF) B-2 Spirit stealth bombers, over one hundred fourth- and fifth-generation fighter aircraft, numerous air-refueling tankers, and a U.S. Navy submarine armed with Tomahawk cruise missiles. As noted by Chairman of the Joint Chiefs of Staff General Dan Caine, the sophisticated attack plan orchestrated by U.S. Central Command received vital support from U.S. European Command, U.S. Transportation Command, U.S. Strategic Command, U.S. Cyber Command, the Joint Staff, U.S. Space Command, and the U.S. Space Force.

U.S. national security space forces played an indispensable role in the planning for and conduct of *Operation Midnight Hammer*. Examples of such key contributions are briefly summarized below.

**Intelligence, Surveillance and Reconnaissance** – Space-based intelligence, surveillance and reconnaissance (ISR) systems were central to the planning and execution of *Operation Midnight Hammer*. ISR satellites, operated by both the DoD and the Intelligence Community (IC) and positioned across multiple orbits, provided valuable insights into the status and disposition of Iranian military forces and facilities, monitored the communications within and among various Iranian military units, determined the operational status and readiness of Iranian air defense systems, and more. Data generated by U.S. Government-owned and -operated space-based ISR systems as well as commercial space-based imagery intelligence (IMINT) systems were key to the successful raid. In addition, electro-optical images taken from space by U.S. commercial imagery companies have helped inform after-action battle damage assessments regarding the extent of destruction to Iranian nuclear facilities at Fordow, Natanz and Isfahan. Likewise, space-based signals intelligence (SIGINT) systems monitored Iranian military communications and provided valuable information on the operational readiness and posture of Iranian air defense radars and other key military equipment.

**Integrated DoD/IC Operational and Intelligence Picture** – The provision of near real-time situational awareness of U.S., allied and adversary forces, disposition, and activity was achieved through seamless transmission and integration of sensor data from and through space to enable the joint picture. U.S. national security space forces across the DoD and IC clearly demonstrated the power of integrated sense-making and awareness.

**Communications** – Space-based communications systems provided critical connectivity to forces engaged in *Operation Midnight Hammer*. As noted by Sandra Erwin of Space News, "During the mission, satellite communications systems enabled near real-time coordination between aircrews, command centers and ISR platforms — a necessity during 18-hour B-2 sorties that required radio silence and stealth over hostile territory."

**Precision Navigation and Timing** – Space-based PNT systems such as the Global Positioning System (GPS) helped guide U.S. bombers and fighters to their designated targets. Such flight routes were carefully planned to maximize security and achieve tactical surprise. GPS was also critical in assuring the pinpoint accuracy and improving the lethality of the Tomahawk sea-launched cruise missiles used to target various Iranian military equipment and facilities and the Massive Ordnance Penetrator bombs used to strike hard and deeply buried targets, especially Iranian nuclear weapons facilities such as Fordow.

**Missile Warning and Theater Missile Defense** – Space-based missile warning systems were used to monitor possible launches of Iranian ballistic missiles targeting U.S. military forces deployed

throughout the region. Following the raid, Iran launched over a dozen missiles against U.S. military forces at the al Udeid Air Base in Qatar. According to various reports, U.S. missile defense systems intercepted 13 of the 14 Iranian missiles launched (the 14th missile flew off course and caused no damage). These highly-effective U.S. theater missile defense operations were directly enabled by U.S. space-based missile warning and tracking systems.

**Adversary Counterspace Operations** – While no attacks on U.S. space systems during *Operation Midnight Hammer* have been reported, U.S. Space Command and U.S. Space Force systems and personnel were no doubt carefully monitoring for adversary counterspace operations in the run-up to and during the raid. For instance, U.S. space operators were on the lookout for Iranian electronic warfare attacks against U.S. on-orbit satellite communications assets and jamming of GPS signals, among other attack vectors. America's space warriors were prepared and empowered to take appropriate action in order to protect and defend critical U.S. space assets.

*Operation Midnight Hammer* clearly demonstrated U.S. military planning and operational prowess and global reach. "Every American involved in this operation performed flawlessly", said Secretary of Defense Hegseth in an after-action briefing. When combined with sustained attacks on Iranian nuclear weapons-related facilities and scientists by the Israeli Defense Force and Israeli intelligence organizations, Tehran's ability to threaten its neighbors has been severely degraded.

*Operation Midnight Hammer* also demonstrated a high degree of cross-Command, cross-domain coordination and jointness. According to Gen. Caine, the operation "was planned and executed across multiple domains and theaters with coordination that reflects our ability to project power globally with speed and precision at the time and place of our choosing." America's space forces are both a vital contributor to and a key enabler of such joint planning and operations.

Yet, one must assume that future conflicts will include significant and sustained attacks on America's "eyes and ears" in space. China and Russia, in particular, possess a broad range of counterspace systems that could significantly degrade or destroy vital U.S. military and commercial satellites and significantly hinder our ability to conduct sophisticated joint military operations. Indeed, both Beijing and Moscow have fielded a wide variety of space and counterspace forces that, had they been used in the run-up to or during *Operation Midnight Hammer*, could have significantly degraded U.S. military effectiveness.

U.S. lawmakers and the American people must take note of the vital role played by U.S. national security space forces, across the DoD and the IC, in modern warfare and provide increased financial resources and other support to assure U.S. space superiority. Preparing for possible future conflicts with America's enemies demands significant additional investments now.

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